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ORIGINAL RESEARCH

Comparison of Cardiovascular Events in Patients Receiving Concomitant Clopidogrel and Proton Pump Inhibitors Classified by CYP2C19 Inhibitory Potency

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BACKGROUND: Proton pump inhibitors (PPIs) may potentially reduce clopidogrel's antiplatelet effect and increase cardiovascular risk. The degree of CYP (cytochrome P450) 2C19 inhibition varies among PPIs. Few studies have evaluated individual PPIs by CYP2C19 inhibition strength across countries. This study aimed to compare the incidence of cardiovascular events between strong CYP2C19-inhibiting potency and weak CYP2C19-inhibiting potency (weak or non-CYP2C19–inhibiting PPIs) in patients receiving clopidogrel.

METHODS: We conducted an international observational cohort study using 14 databases from the United States, South Korea, and Taiwan. We included patients aged ≥18 years who received clopidogrel with PPIs from 1985 to 2023. PPIs were classified into strong CYP2C19-inhibiting PPIs and weak or non-CYP2C19–inhibiting PPIs based on CYP2C19 inhibition. We compared the hazard ratios and 95% CIs for major adverse cardiovascular events, including myocardial infarction, stroke, and cardiovascular mortality, using the Cox proportional hazards model after 1:1 propensity score matching. Secondary outcomes included cardiovascular mortality, myocardial infarction, stroke, and all-cause mortality. A random-effects model calculated pooled hazard ratios and 95% CIs. Only databases meeting all diagnostic criteria were included in the meta-analysis.

RESULTS: Large-scale propensity score matching identified 166 005 patient pairs. During the 365-day follow-up, the risk of major adverse cardiovascular events did not differ significantly between patients receiving clopidogrel plus strong CYP2C19-inhibiting PPIs and those receiving clopidogrel plus weak or non-CYP2C19–inhibiting PPIs (17.63 per 1000 person-years versus 16.82 per 1000 person-years; calibrated hazard ratio, 1.00 [95% CI, 0.79–1.26]). No significant difference was observed in the risk of secondary outcomes (calibrated hazard ratio, cardiovascular mortality 1.10 [95% CI, 0.87–1.39], myocardial infarction 0.98 [95% CI, 0.81–1.19], stroke 1.05 [95% CI, 0.87–1.27], and all-cause mortality 1.18 [95% CI, 0.93–1.50]).

CONCLUSIONS: Concomitant use of clopidogrel and strong CYP2C19-inhibiting PPIs was not associated with a higher cardiovascular risk compared with concomitant use of clopidogrel and weak or non-CYP2C19–inhibiting PPIs. This large-scale study does not support the clinical significance of potential interactions between PPIs and clopidogrel.

Key Words: clopidogrel ■ mortality ■ myocardial infarction ■ proton pump inhibitors ■ risk factors

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Clopidogrel is a cornerstone of antiplatelet therapy for preventing ischemic cardiovascular events, but its use is associated with an increased bleeding risk. Recent evidence suggests that bleeding events carry a prognostic significance comparable to that of postdischarge myocardial infarction (MI) in patients with

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Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/ATVBAHA.125.323399.
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Nonstandard Abbreviations and Acronyms

ACS	acute coronary syndrome
COGENT	Clopidogrel and the Optimization of Gastrointestinal Events Trial
CYP	cytochrome P450
EASE	expected absolute systematic error
H2RA	histamine-2 receptor antagonist
HR	hazard ratio
IRB	institutional review board
MACE	major adverse cardiovascular events
MI	myocardial infarction
PPI	proton pump inhibitors
PS	propensity score
sPPI	strong CYP2C19-inhibiting proton pump inhibitor
wPPI	weak or non-CYP2C19–inhibiting proton pump inhibitor

What Are the Clinical Implications?

This large-scale, multinational observational study provides important real-world evidence regarding the clinical safety of concomitant proton pump inhibitor use in patients treated with clopidogrel. Despite longstanding concerns that strong CYP (cytochrome P450) 2C19–inhibiting proton pump inhibitors may attenuate clopidogrel's antiplatelet effect and increase cardiovascular risk, our findings demonstrate no significant difference in major adverse cardiovascular events between patients receiving clopidogrel with strong versus weak CYP2C19-inhibiting proton pump inhibitors across diverse healthcare systems. These results suggest that pharmacokinetic interactions observed in experimental settings do not necessarily translate into clinically meaningful harm at the population level. From a clinical perspective, this supports greater flexibility in proton pump inhibitor selection for patients receiving clopidogrel, allowing clinicians to prioritize gastrointestinal bleeding prevention without compromising cardiovascular safety. This is particularly relevant for high-risk populations in whom gastroprotection is essential. Future research should integrate clinical outcomes with pharmacogenetic and mechanistic data to better identify patient subgroups in whom such interactions may be clinically relevant, ultimately enabling more individualized and evidence-based prescribing decisions.

acute coronary syndrome (ACS).[1] Gastrointestinal bleeding, the most common bleeding complication,[2] can be effectively prevented by proton pump inhibitors (PPIs).[3] However, concerns persist about potential drug interactions between PPIs and clopidogrel via CYP (cytochrome P450) 2C19, the enzyme responsible for converting clopidogrel into its active metabolite.

The strength of CYP2C19 inhibition varies substantially among PPIs. Pharmacodynamic studies have demonstrated that PPIs with strong CYP2C19-inhibiting potency (sPPIs), such as omeprazole, significantly reduce the antiplatelet effect of clopidogrel compared with those with weak or non-CYP2C19–inhibiting PPIs (wPPIs), such as pantoprazole and rabeprazole.[4,5] The US Food and Drug Administration and the European Medicines Agency issued a cautionary statement on the concomitant use of clopidogrel and omeprazole or esomeprazole.[6,7] The recent European Society of Cardiology guideline recommends against prescribing omeprazole and esomeprazole to patients receiving clopidogrel.[8]

Nevertheless, strong CYP2C19 inhibitors such as omeprazole and esomeprazole remain widely prescribed in real-world clinical practice.[9] These agents provide potent and consistent acid suppression, with extensive evidence supporting their efficacy and safety for gastroprotection, particularly in patients at high risk of gastrointestinal bleeding.[10–13] In such patients, avoiding potent PPIs solely due to potential pharmacokinetic interactions may paradoxically increase the risk of serious gastrointestinal complications. From a gastroenterological perspective, omeprazole and esomeprazole therefore remain essential therapeutic options, making it clinically important to clarify whether their concomitant use with clopidogrel truly increases cardiovascular risk.

Despite ongoing debate, the clinical significance of differential PPI effects on cardiovascular outcomes remains largely unexplored. Most previous studies used nonusers as a control group,[14] which may introduce bias in observational studies, as patients prescribed PPIs often have more comorbidities and a higher baseline risk for adverse outcomes than those who do not receive PPIs.[15] This imbalance can exaggerate the association between PPI use and adverse clinical events. Therefore, we conducted an international cohort study using data from the United States, South Korea, and Taiwan to compare the risk of major adverse cardiovascular events (MACEs) between sPPIs and wPPIs when added to established clopidogrel therapy.

MATERIALS AND METHODS

Data Sources

Electronic medical records and administrative claims data converted to the Observational Medical Outcomes Partnership Common Data Model were used from 14 data sources. Details of the included databases are provided in Table S1. All databases consisted of deidentified, patient-level electronic health records transformed into a common data model–standardized vocabulary using the same analysis program.[16–18] This study followed the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines.

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This study was approved by the institutional review board (IRB) of the Severance Hospital (IRB No. 4-2023-0200), Kangdong Sacred Heart Hospital (IRB No. 2021-07-011-001), and all data partners in the United States and Taiwan (N202503060). The other 7 hospitals in Korea joined the Research Border Free Zone of the common data model data network, which recognizes the IRB approval of research institutions and exempts IRB approval at each institution. The requirement for informed consent was waived because of the retrospective nature of the study. This study was registered in the European Union electronic register of postauthorization studies (EU PAS Register; registration number: EUPAS1000000203).

Cohort Design and Study Population

We conducted a population-based, retrospective cohort study. We enrolled patients aged ≥18 years who received clopidogrel at the time of PPI initiation and had an observation period of at least 1 year before the index date. The study population included only patients who received clopidogrel for >30 consecutive days. The first prescription date of a PPI added on top of this ongoing clopidogrel treatment was defined as the index date. Patients with a history of MI or stroke within 1 year before the index date were excluded to avoid misclassification when estimating the incidence rates of the primary outcome.

PPIs were classified according to their CYP2C19 inhibitory potency and metabolic characteristics based on previous studies.[4,11,19–23] Esomeprazole and omeprazole were categorized as sPPIs because they act as irreversible or quasi-irreversible metabolism-dependent inhibitors of CYP2C19, whereas dexlansoprazole, lansoprazole, pantoprazole, and rabeprazole were defined as wPPIs. wPPIs were defined as active comparators because they have similar indications and a weak or no effect on CYP2C19 compared with observations in sPPIs.[19,24]

The index date was defined as the first prescription date of a PPI while continuing clopidogrel therapy. Continuous drug exposure was inferred by allowing gaps of <30 days between prescription records based on our previous study.[25] A schematic description of the study design is shown in Figure S1.

Outcome and Covariates

The primary outcome was MACE, which included cardiovascular mortality and hospitalization or emergency department visits for MI and stroke within 365 days after the index date (Table S2).[26] Secondary outcomes were defined as the individual components of MACE (cardiovascular mortality, MI, and stroke) and all-cause mortality (Table S2). The study population was censored at the earliest occurrence of death, the last date of the observation period, a 30-day treatment gap indicating PPI discontinuation, use of comparator drugs in the target group, or use of target drugs in the comparator group.

Covariates were used to match the target and comparator cohorts and included patient demographics (age, sex, and race), index year, prior comorbidities and medication use, historical procedures and measurements, device exposures within 30 and 365 days before the index date, and the Charlson Comorbidity Index over 365 days before the index date.

Statistical Analysis

Proportions and standardized mean differences of baseline patient characteristics were calculated. We used 1:1 propensity score (PS) matching to adjust for potential confounders and balance the target and comparator cohorts, using large-scale logistic regression models with covariates from the 12 months preceding the index date. A PS model was constructed for each target-comparator pair.

The primary analysis used an intention-to-treat analysis up to 365 days of the index date. Sensitivity analyses were performed using different comparators, time-at-risk windows, PS methods, and populations. First, comparator drugs: histamine-2 receptor antagonists (H2RAs), which have a weak or no effect on CYP2C19, were added as comparators. H2RAs included cimetidine, famotidine, lafutidine, nizatidine, ranitidine, and roxatidine. In addition, we compared the overall clopidogrel+overall PPI population (including both sPPIs and wPPIs) with the clopidogrel+H2RA population. Second, time-at-risk windows: 4 different risk periods were defined: (1) 30 days after PPI treatment initiation; (2) 365 days after PPI treatment initiation; (3) intention-to-treat: patients were followed until the end of the observation period; (4) during treatment: patients were followed until discontinuation of either PPIs or clopidogrel. Third, PS models: we applied 3 PS methods: 1:1 PS matching, 1:variable-ratio PS matching, and PS stratification (10 strata). Fourth, study population subgroups: 8 population pairs were analyzed: (1) concomitant use of clopidogrel and sPPIs versus clopidogrel and wPPIs; (2) concomitant use of clopidogrel and sPPIs versus clopidogrel and H2RAs; (3) concomitant use of clopidogrel and sPPIs in patients with ACS versus clopidogrel and wPPIs in patients with ACS; and (4) concomitant use of clopidogrel and sPPIs in patients without previous gastrointestinal bleeding versus clopidogrel and wPPIs in patients without previous gastrointestinal bleeding. The presence of ACS and gastrointestinal bleeding was determined based on diagnoses recorded within 1 year before the index date. Each of these 4 pairs was further stratified based on the presence or absence of a history of MI or stroke. Subgroup analyses were conducted by age (<65 versus ≥65 years), sex, region (Asia versus non-Asia), with versus without diabetes, and with versus without aspirin among data sources that met the predefined diagnostic criteria, excluding the Kyung Hee University Medical Center database.

Cumulative incidence of the primary outcome was estimated using the Kaplan-Meier curve, and differences between groups were assessed using the log-rank test. Calibrated hazard ratios (HRs) and 95% CIs were calculated using the Cox proportional hazards model.[27] Statistical significance was defined as a 2-sided $P<0.05$. A random-effects meta-analysis was performed, and heterogeneity was assessed using the I^2 statistic.[28,29] Database construction and statistical analyses were performed using R software version 4.1.2 (R Foundation for Statistical Computing), using open-source statistical packages from the Observational Health Data Sciences and Informatics HADES software collection.[30]

Predefined Diagnostics

Three prespecified objective study diagnostic rules were implemented to ensure reliable analysis.[18] The rules included: (1) assessment of baseline characteristics to check covariate

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balance after PS methods; (2) preference score distributions (a transformation of PS) to evaluate empirical equipoise; and (3) expected absolute systematic error (EASE) to assess residual bias. PS adjustment was used to reduce the effects of confounding factors by balancing the confounder proportions. Covariate balance was assessed using the standardized mean difference, with values <0.25 indicating adequate balance.[31,32] In the 2 treatment groups, clinical equipoise was measured by the proportion of overlap in the treatment groups' preference score distributions. We set indifference in treatment selection between the target and comparator cohorts if >30% of participants in clinical equipoise had preference scores in the range of 0.3 to 0.7.[33] For each analysis, we estimated the EASE to check for systematic error.[34] The EASE score is a measurement tool to determine how much the observed risk differs from the actual risk of 257 negative control outcomes (Table S3). If systematic errors exist, the EASE score is high. The threshold for the EASE score was set at <0.25. The prespecified study diagnostic rules were considered satisfied when all 3 conditions were met, and only databases that met all rules were included in each meta-analysis.

Data Availability

This study was conducted using a distributed research network based on a common data model, in which patient-level data remain under the control of the respective data partners. The patient-level data used in this study were not shared due to concerns regarding patient privacy. The study protocol and open, executable source code used in this study are available at https://github.com/ohdsi-studies/DdiPpiClo.

RESULTS

Cohort Selection and Characteristics

Among the 14 databases, 3 were from the United States, 1 was from Taiwan, and 10 were from South Korea. Between January 1985 and December 2023, 479,041 new eligible users of clopidogrel and sPPIs (239 427 patients) or wPPIs (239 614 patients) were identified (Figure 1). Among the study population, women comprised 45% of sPPI users and 44% of wPPI users (Table). Among 168 738 PS-matched pairs, the prevalence of comorbidities was as follows: hypertension (78%), ischemic heart disease (56%), gastroesophageal reflux disease (50%), and diabetes (37%). In addition, lipid-modifying agents were prescribed to 75% of patients in both groups. The prescription rate of aspirin was 40% in the clopidogrel+sPPIs and 39% in the clopidogrel+wPPIs. Comparing clopidogrel+sPPIs with clopidogrel+H2RA, the prescription rate of aspirin was also comparable between the 2 groups (39% versus 42% before PS matching). After PS matching, among 168 738 matched pairs, the standardized mean differences for all baseline characteristics between sPPI and wPPI users were <0.1 within each data source, indicating better balance than that before matching (Figures S2 and S3).

Results From Study Diagnostics Rules

The diagnostic results are shown in Table S4. After PS matching, the maximum values of absolute standardized

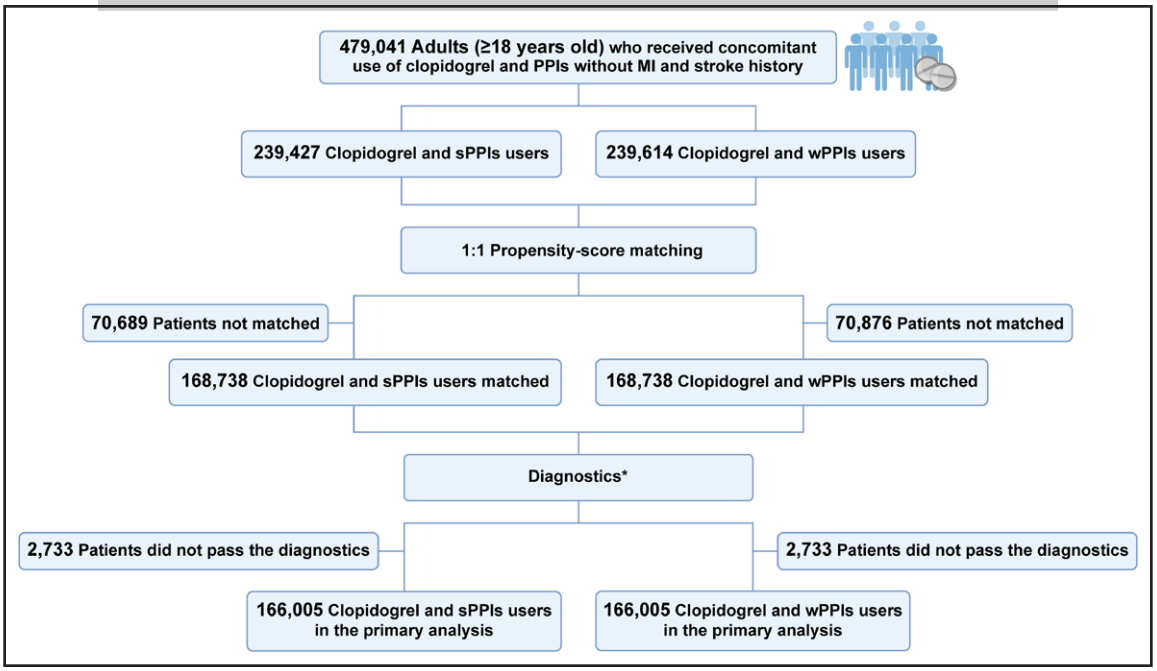

Figure 1. Study flowchart.
CYP indicates cytochrome P450; sPPI, strong CYP2C19-inhibiting proton pump inhibitor; and wPPIs, weak or non-CYP2C19–inhibiting proton pump inhibitor. *The objective study diagnostics included (1) assessment of baseline characteristics to check covariate balance after propensity score (PS) methods; (2) preference score distributions (a transformation of PS) to evaluate empirical equipoise; and (3) expected absolute systematic error to assess residual bias. All 3 diagnostic rules needed to be satisfied for the analysis to be included in the analysis.

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Table. Baseline Characteristics of Patients Before and After Propensity Score Matching Between Clopidogrel+sPPIs and Clopidogrel+wPPIs

Characteristic	Before propensity score matching			After propensity score matching		
	Clopidogrel+sPPIs (N=239 427)	Clopidogrel+wPPIs (N=239 614)	SMD	Clopidogrel+sPPIs (N=168 738)	Clopidogrel+wPPIs (N=168 738)	SMD
Age group, y, %						
<55	11.3	11.4	0.00	11.5	11.3	0.00
55–59	12.0	11.6	0.01	11.9	11.9	0.00
60–64	17.2	16.9	0.01	17.5	17.5	0.00
65–69	15.1	15.0	0.00	15.1	15.1	0.00
70–74	15.7	16.0	−0.01	15.7	15.7	0.00
75–79	14.1	14.3	0.00	14.1	14.1	0.00
80–84	9.4	9.6	−0.01	9.2	9.3	0.00
85+	5.0	5.3	−0.01	5.1	5.1	0.00
Sex: women, %	44.6	44.2	0.01	41.6	41.8	0.00
Medical history, %						
Hyperlipidemia	81.7	80.7	0.03	82.0	81.9	0.00
Hypertensive disorder	77.4	77.0	0.01	77.8	77.6	0.00
Acute respiratory disease	59.7	59.1	0.01	63.0	62.7	0.01
Ischemic heart disease	56.4	57.1	−0.01	56.4	56.1	0.01
Gastroesophageal reflux disease	46.8	47.4	−0.01	50.0	49.4	0.01
Diabetes	40.1	40.3	0.00	37.0	36.8	0.00
Peripheral vascular disease	35.6	34.4	0.03	35.5	35.3	0.00
Osteoarthritis	31.8	28.2	0.08	28.6	28.6	0.00
Cerebrovascular disease	25.6	25.3	0.01	25.5	25.5	0.00
Heart failure	18.0	18.0	0.00	17.9	17.8	0.00
Depressive disorder	17.5	17.5	0.00	17.5	17.3	0.01
Renal impairment	8.2	9.4	−0.05	8.7	8.7	0.00
Dementia	7.4	7.7	−0.01	7.5	7.5	0.00
Rheumatoid arthritis	4.9	4.4	0.03	4.5	4.5	0.00
Medication use, %						
Anti-inflammatory and antirheumatic products	84.5	81.9	0.07	81.6	81.6	0.00
Drugs for acid-related disorders	83.9	84.3	−0.01	84.1	83.9	0.00
Lipid-modifying agents	75.1	75.4	−0.01	75.2	75.2	0.00
Agents acting on the renin-angiotensin system	61.5	62.7	−0.02	61.7	61.8	0.00
Angiotensin II antagonists	54.1	54.9	−0.01	54.3	54.4	0.00
Calcium channel blockers	52.5	53.3	−0.02	52.6	52.5	0.00
β-Blockers	43.6	45.4	−0.04	44.1	44.0	0.00
Aspirin	39.2	41.6	−0.05	39.5	39.4	0.00
Diuretics	38.4	40.6	−0.04	38.9	38.8	0.00
Warfarin	1.5	1.8	−0.03	1.6	1.6	0.00

CYP indicates cytochrome P450; SMD, standardized mean difference; sPPI, strong CYP2C19-inhibiting proton pump inhibitor; and wPPI, weak or non-CYP2C19–inhibiting proton pump inhibitor.

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mean difference for the 8 databases (167 648 pairs; database names: Health Insurance Review and Assessment, IBM Health MarketScan Commercial Claims and Encounters Database, Columbia University Medical Center, Ajou University School of Medicine, Gyeongsang National University Hospital, Kyung Hee University Medical Center, Pusan National University Hospital, and Yonsei University Health System) were lower than 0.25. For preference score distribution overlap (clinical equipoise), all databases except Taipei Medical University Clinical Research Database met the threshold value (Table S4; Figures S4 and S5). The

range of preference score overlap in the passed data sources was 0.47 to 0.93. The EASE values, representing systematic bias, were <0.25 in 6 databases. The range of EASE was 0.01 to 0.19 in these 6 databases. Among the 168 783 pairs, there were 5 databases (166 005 pairs; database names: Health Insurance Review and Assessment, IBM Health MarketScan Commercial Claims and Encounters Database, Columbia University Medical Center, Kyung Hee University Medical Center, and Yonsei University Health System) where all 3 study diagnostic rules were passed. Only 5 databases that passed all objective study diagnostics were included in the meta-analysis.

Association of Concomitant Use of Clopidogrel and sPPIs With Cardiovascular Risk

During the 365 days after the index date, the MACE incidence rates were 17.63 per 1000 person-years (2822 patients) in the clopidogrel plus sPPIs group and 16.82 per 1000 person-years (2696 patients) in the clopidogrel plus wPPIs group (Figure 2). The incidence rates and cumulative incidence curves of the primary outcome in each database are shown in Figures 2 and 3, respectively. The risk of MACE was not statistically different between clopidogrel plus sPPIs and clopidogrel plus wPPIs (calibrated HR, 1.00 [95% CI, 0.79–1.26]; Figure 2). In the Kaplan-Meier curves, a statistically significant difference in cumulative MACE incidence between the 2 groups was observed in the IBM Health MarketScan Commercial Claims and Encounters Database, whereas no significant differences were detected in the remaining databases (Columbia University Medical Center, Health Insurance Review and Assessment, Kyung Hee University Medical Center, and Yonsei University Health System; Figure 3). Among the 120 negative control outcomes, 117 (97.5%) in the clopidogrel+sPPIs group versus the clopidogrel+wPPIs group had meta-analytic HRs with 95% CIs covering one after calibration (Figure S6).

There were no significant differences in the risk of cardiovascular mortality (incidence rate at 1 year, 11.74 per 1000 person-years in the clopidogrel+sPPIs group versus 11.36 per 1000 person-years in the clopidogrel+wPPIs group; calibrated HR, 1.10 [95% CI, 0.87–1.39]), MI (<7.42 per 1000 person-years in the clopidogrel+sPPIs group versus 7.68 per 1000 person-years in the clopidogrel+wPPIs group; calibrated HR, 0.98 [95% CI, 0.81–1.19]), stroke (<13.06 per 1000 person-years in the clopidogrel+sPPIs group versus 13.20 per 1000 person-years in the clopidogrel+wPPIs group; calibrated HR, 1.05 [95% CI, 0.87–1.27]), or all-cause mortality (30.66 per 1000 person-years in the clopidogrel+sPPIs group versus 30.89 per 1000 person-years in the clopidogrel+wPPIs group; calibrated HR, 1.18 [95% CI, 0.93–1.50]; Figure 4).

Sensitivity Analyses and Subgroup Analyses

The results of the sensitivity analyses using diverse time windows and PS models are shown in Table S5. The risk of MACE was slightly higher in the clopidogrel+sPPIs group than in the clopidogrel+wPPIs group in the 30-day posttreatment analysis (calibrated HR, 1.19 [95% CI, 1.01–1.40]) and the on-treatment analysis (calibrated HR, 1.32 [95% CI, 1.10–1.59]). However, no significant difference was observed in the intention-to-treat analysis (calibrated HR, 1.06 [95% CI, 0.91–1.23]). Concomitant use of clopidogrel and sPPIs, compared with concomitant use of clopidogrel and wPPIs, was not associated with an increased risk of MACE using 1:variable-ratio PS matching (calibrated HR, 0.93 [95% CI, 0.69–1.24]) or PS stratification (calibrated HR, 1.00 [95% CI, 0.83–1.21]).

The results of the sensitivity analyses for each outcome across the various target-comparator cohort pairs are shown in Table S6. The results using H2RA as an alternative active comparator were consistent (calibrated HR for MACE, 0.90 [95% CI, 0.60–1.34]) with those

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Data Source	Incidence Rate Clo + sPPIs	Incidence Rate Clo + wPPIs	Hazard Ratio (95% CI)
CCAE	25.25	20.41	1.25 (1.07 - 1.47)
CUMC	18.24	27.46	0.66 (0.37 - 1.17)
HIRA	17.09	16.50	1.04 (0.99 - 1.11)
KHMC	10.63	14.78	0.71 (0.28 - 1.79)
YUHS	5.12	8.66	0.51 (0.20 - 1.31)
Meta-analysis (I^2 = 54.1%)	**17.63**	**16.82**	**1.00 (0.79 - 1.26)**

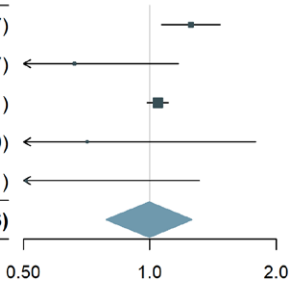

Figure 2. Comparative risks of the primary outcome (major adverse cardiovascular events) between clopidogrel (Clo)+strong CYP (cytochrome P450) 2C19–inhibiting proton pump inhibitors (sPPIs) and Clo+weak or non-CYP2C19–inhibiting proton pump inhibitors (wPPIs) in meta-analysis.
CCAE indicates IBM Health MarketScan Commercial Claims and Encounters Database; CUMC, Columbia University Medical Center; HIRA, Health Insurance Review and Assessment; KHMC, Kyung Hee University Medical Center; and YUHS, Yonsei University Health System.

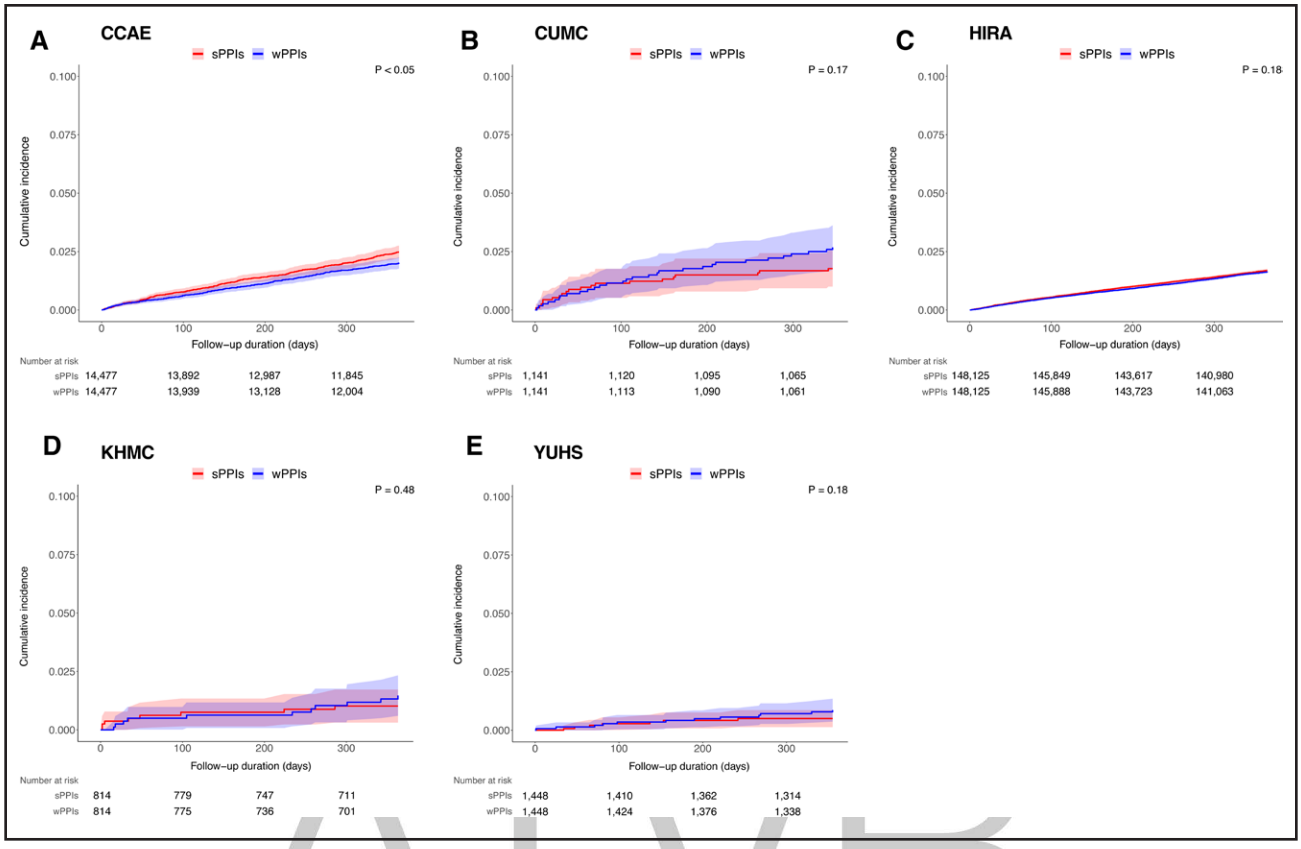

Figure 3. Kaplan-Meier plots for risk of the primary outcome (major adverse cardiovascular events) between clopidogrel (Clo)+strong CYP (cytochrome P450) 2C19-inhibiting proton pump inhibitors (sPPIs) vs Clo+weak or non-CYP2C19–inhibiting proton pump inhibitors (wPPIs).
CCAE indicates IBM Health MarketScan Commercial Claims and Encounters Database; CUMC, Columbia University Medical Center; HIRA, Health Insurance Review and Assessment; KHMC, Kyung Hee University Medical Center; and YUHS, Yonsei University Health System.

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of the primary analysis. When comparing the overall clopidogrel+PPI population with the clopidogrel+H2RA group, the risk of MACE was also not significantly different between the 2 groups (meta-analytic HR, 0.99 [95% CI, 0.87–1.13]). There was no significantly higher MACE risk in patients with ACS (calibrated HR,

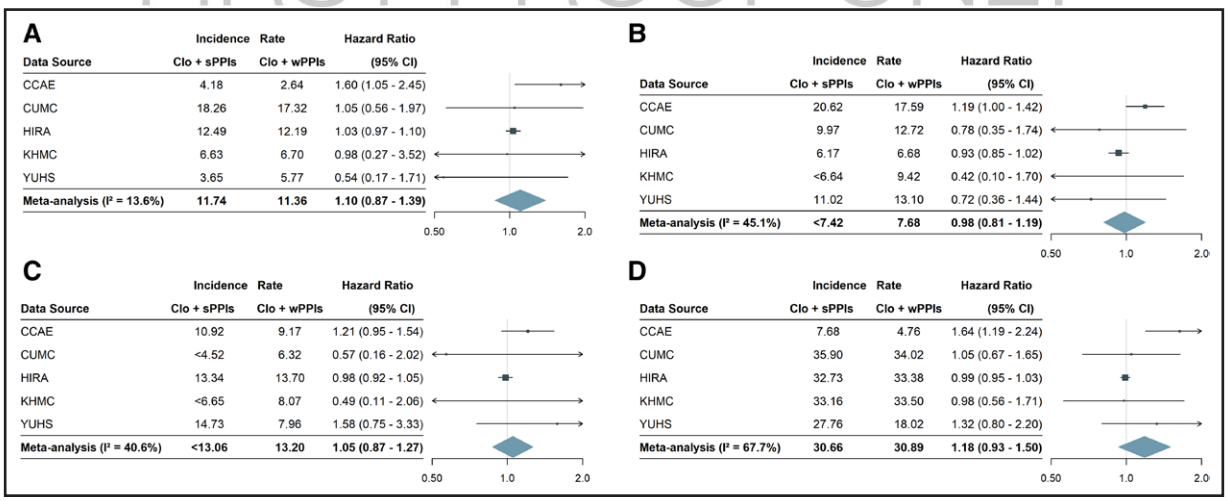

Figure 4. Comparative risks of the secondary outcome (cardiovascular mortality, myocardial infarction, stroke, and all-cause mortality) between clopidogrel (Clo)+strong CYP (cytochrome P450) 2C19-inhibiting proton pump inhibitors (sPPIs) and Clo+weak or non-CYP2C19–inhibiting proton pump inhibitors (wPPIs) in meta-analysis.
CCAE indicates IBM Health MarketScan Commercial Claims and Encounters Database; CUMC, Columbia University Medical Center; HIRA, Health Insurance Review and Assessment; KHMC, Kyung Hee University Medical Center; and YUHS, Yonsei University Health System.

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0.99 [95% CI, 0.89–1.10]) or in those without previous gastrointestinal bleeding (calibrated HR, 1.03 [95% CI, 0.90–1.18]). In addition, we confirmed that there was no association with cardiovascular events when comparing the cohorts, even after including individuals with a history of MI and stroke.

The results of the subgroup analyses are shown in Figure S7. Across all subgroups stratified by age, sex, region, diabetes comorbidity, and concomitant aspirin use, the risk of MACE did not differ between patients treated with clopidogrel+sPPIs and those treated with clopidogrel+wPPIs.

DISCUSSION

This study evaluated the comparative cardiovascular risk associated with the concomitant use of clopidogrel and sPPIs versus clopidogrel and wPPIs across 14 databases, including populations from the United States, Taiwan, and South Korea. This study analyzes the clinical interaction between clopidogrel and sPPIs versus that between clopidogrel and wPPIs across countries. In the large-scale observational study, the meta-analysis found no significant difference in the risk of MACE between the 2 groups (calibrated HR, 1.00 [95% CI, 0.79–1.26]) or in secondary outcomes. Sensitivity analyses across various time windows and comparison groups confirmed these findings. However, a short-term effect was observed, with a statistically significant increase in MACE risk in the early exposure windows.

The potential interaction with pharmacodynamic consequences has been reported between PPIs and clopidogrel in several studies.[20,35] One proposed mechanism suggests that drug absorption is more efficient in an acidic environment; therefore, the use of PPIs or other acid-reducing agents may elevate gastric pH, potentially hindering or delaying the absorption process.[36] A more widely recognized mechanism suggests that PPIs interfere with the activation of clopidogrel by inhibiting CYP2C19, a key enzyme involved in its metabolic conversion to active metabolites.[37] Because PPIs, particularly omeprazole and esomeprazole, also rely on CYP2C19 for their own metabolism, their concurrent use with clopidogrel may lead to competitive inhibition at the enzymatic level. This interaction can reduce the bioavailability of clopidogrel's active metabolite, potentially diminishing its antiplatelet efficacy.[15,35–37]

Omeprazole and esomeprazole exhibit irreversible or quasi-irreversible, metabolism-dependent inhibition of CYP2C19, whereas other PPIs, such as lansoprazole, act as mild inhibitors, and pantoprazole and rabeprazole exert minimal or no inhibition of this hepatic enzyme.[11,38,39] Several pharmacokinetic studies and randomized clinical trials have demonstrated that omeprazole and esomeprazole may attenuate the antiplatelet activity of clopidogrel by competitively inhibiting CYP2C19-mediated biotransformation to its active metabolite.[4,11,20–23] Consequently, both the European Medicines Agency and US Food and Drug Administration issued warnings regarding the potential interaction of clopidogrel with omeprazole or esomeprazole.[6,7] As a precaution, they advised against the concurrent use of these medications unless there was a compelling clinical need. However, findings from the COGENT (Clopidogrel and the Optimization of Gastrointestinal Events Trial), the only large-scale randomized controlled trial evaluating the effects of PPIs in patients receiving dual antiplatelet therapy, showed that omeprazole significantly reduced gastrointestinal events without increasing cardiovascular risks.[40] Nevertheless, the aforementioned trial specifically investigated omeprazole and was not adequately powered to definitively rule out a potential interaction between PPI use and cardiovascular outcomes.

Given these concerns, the 2023 European guideline advised against the use of sPPIs for gastric protection in patients with diabetes receiving clopidogrel for cardiovascular disease management.[8] This recommendation was based on a meta-analysis reporting a significantly increased risk of MACE with omeprazole (odds ratio, 1.40 [95% CI, 1.15–1.70]) and esomeprazole (odds ratio, 1.59 [95% CI, 1.29–1.95]).[8,14] However, a closer examination of the meta-analysis reveals several limitations. Some observational studies included in the analysis lacked robust methodologies, such as PS analysis, leading to baseline characteristic differences and potential uncontrolled confounding.[41] In addition, in observational studies, PPI users often had more comorbidities and higher baseline risks compared with those not receiving PPIs, which may have introduced bias and exaggerated adverse outcomes.[15] However, all observational studies in the meta-analysis used nonuse as a control group. To mitigate this, we selected weak competitive inhibitors, such as wPPIs and H2RAs, as active comparators. Third, the odds ratios were derived from unadjusted event counts rather than the risk values from original studies. Finally, combining randomized controlled trials with observational studies in a single meta-analysis may have introduced a potential source of bias.

Strengths and Limitations

Our study has several methodological strengths. To enhance the methodological rigor and reliability of our findings, we applied 3 prespecified diagnostic criteria when selecting databases for meta-analysis: covariate balance assessment, clinical equipoise evaluation, and residual bias estimation. Only databases that satisfied all 3 conditions were included: (1) ensuring well-balanced covariates, (2) maintaining sufficient overlap in preference score distributions (clinical equipoise), and (3) minimizing residual bias, as assessed through an extensive

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range of negative control outcomes. By restricting our analysis to databases that met these strict criteria, we reduced confounding, maintained equipoise, and controlled for systematic error, thereby strengthening the validity and robustness of our meta-analysis compared with previous studies.[31,33,34] Defining decision thresholds in advance mitigates biases arising from the retrospective justification of diagnostic shortcomings or assumption breaches.[42] Moreover, we included a large number of study participants from 14 data sources that were converted to the Observational Medical Outcomes Partnership Common Data Model, which reduces the potential for bias specific to individual databases.

Despite its strengths, this observational cohort study has some limitations. First, prescriptions did not account for actual patient use or potential over-the-counter purchases. However, only certain PPIs, such as omeprazole and lansoprazole, have been approved for over-the-counter use.[23] Furthermore, PPIs are not available as over-the-counter drugs in South Korea. Second, although we used a previously validated outcome definition, using diagnostic codes in the operational definition of the outcome may result in potential misclassification bias.[43] Third, the potential for residual confounding factors cannot be excluded from this observational study. To enhance comparability between the exposure groups, we implemented a large-scale PS matching. Fourth, we did not assess genetic polymorphisms because CYP2C19 genotype information was unavailable. Therefore, the applicability of our findings to patients carrying CYP2C19 loss-of-function alleles should be interpreted with caution. Although the IBM Health MarketScan Commercial Claims and Encounters Database showed a numerically higher risk of MACE among clopidogrel–sPPI users, no significant difference in treatment effects was observed between Asian and non-Asian databases, and this pattern was not reproduced in the overall meta-analysis or regional subgroup analyses. However, given that CYP2C19 allele frequencies are known to differ substantially across populations, a genotype-related effect cannot be completely excluded. Although residual confounding cannot be fully ruled out, the large-scale PS model used in this study, which incorporated thousands of baseline clinical and medication covariates, may have partly mitigated unmeasured or indirectly measured confounding through correlated covariates, as demonstrated in recent methodological research.[44] The potential influence of CYP2C19 genetic polymorphisms on the clopidogrel and PPI interaction warrants further investigation.

Conclusions

When a rigorous study design with predefined diagnostic rules was used to assess bias, there was no difference in the incidence of major cardiovascular events between sPPIs and wPPIs in patients taking clopidogrel. This large-scale study did not support the clinical significance of potential interactions between PPIs and clopidogrel. Still, a transient early increase in risk was noted, and enhanced clinical attention during the early phase of sPPI initiation in clopidogrel users may be reasonable.

ARTICLE INFORMATION

Received July 22, 2025; accepted February 9, 2026.

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Acknowledgments

The authors thank Jaehyun Shin, MFA (Research Associate, Department of Biomedical Systems Informatics, Yonsei University College of Medicine) and MID (Medical Illustration & Design, a member of the Medical Research Support Services at Yonsei University College of Medicine) for providing support with medical illustration.

Author Contributions

Seonji Kim, S.C. You, and S.I. Seo contributed to concept and design. Seonji Kim, S.C. You, and S.I. Seo drafted the manuscript. Seonji Kim contributed to other editorials/conceptuals. G. Hripcsak, J.C. Hsu, P. Nagy, W.G. Shin, S.C. You, and S.I. Seo contributed to critical review of the manuscript for important intellectual content. Seonji Kim, J. Park, H.Y. Chen, C.H. Setiawan, H. Lee, B. Martin, K.J. Lee, Subin Kim, P.-A. Nguyen, P. Thanh-Phuc, M.S. Muhtar, and M.-H. Hsu contributed to the statistical analysis. S.C. You and S.I. Seo obtained funding and supervised the study. All authors contributed to the acquisition, analysis, or interpretation of data.

Sources of Funding

This work was supported by the Korean College of Helicobacter and Upper Gastrointestinal Research Foundation Grant (KCHUGR 202102003) and the National Research Foundation of Korea grant funded by the Korea government (MSIT; RS-2024-00341426).

Disclosures

S.C. You reports grants from Daiichi Sankyo. He is a coinventor of granted Korean patents DP-2023-1223 and DP-2023-0920 and has pending Patent Applications DP-2024-0909, DP-2024-0908, DP-2022-1658, DP-2022-1478, and DP-2022-1365, unrelated to the current work. He is also the Chief Executive Officer of PHI Digital Healthcare. The other authors report no conflicts of interest.

Supplemental Material

Tables S1–S6
Figures S1–S7

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